

Profile

Name Thevathayarajh Thayananthan

Label Ph.D. Candidate, Agricultural Engineering — University of Georgia

Summary Agricultural robotics and computer vision. I build vision-guided autonomous systems for selective harvesting and food processing, from simulation through field deployment. Currently searching for new positions.

Education

2025 - PRESENT

Ph.D.

University of Georgia (UGA)

Agricultural Engineering

2023 - 2025

Ph.D. coursework and research (transferred to UGA with advisor)

Mississippi State University

Biosystems Engineering

2019 - 2022

M.Sc.

University of Peradeniya

Electrical and Electronics Engineering

2014 - 2018

B.Tech. (Hons.)

Uva Wellassa University of Sri Lanka

Science and Technology (Mechatronics)

Research Appointments

2025 - PRESENT

Graduate Research Assistant

Sensing & Automation in Agri-Systems (SAAS) Lab, University of Georgia

- Perception-driven manipulator control in ROS and Python for a dual-side robotic cotton picker, integrating two ZED2i stereo cameras, a UR5e manipulator, and a custom end-effector
- Benchmarked 34 detection and segmentation models (30 YOLO variants v8–v26 and four RF-DETR variants) on a 1,252-image, four-class annotated field dataset
- Built CottonID, a full-stack web application (Next.js, FastAPI, SQLite) that converts field imagery into boll counts, yield estimates, and defoliation-timing recommendations for growers
- Manage equipment, operations, data infrastructure, and safety for a three-researcher agricultural robotics laboratory

2023 - 2025

Graduate Research Assistant

Department of Agricultural and Biological Engineering, Mississippi State University

- Built and open-sourced CottonSim: a ROS/Gazebo simulator and vision-guided navigation stack for autonomous cotton picking (100% GPS-guided field-traversal completion; 96.7% map-based)
- Multi-ripeness blackberry detection for soft robotic harvesting; benchmarked nine YOLO architectures on a two-season, 1,086-image field dataset
- Deep learning semantic segmentation for automated catfish processing, with USDA-ARS (SegFormer-B5: 89.2% mIoU)
- Named as an inventor on a university technology disclosure for an autonomous vision-guided cotton-harvesting system

2024

Graduate Teaching Assistant

Department of Agricultural and Biological Engineering, Mississippi State University

- ABE 4423 / ABE 6423 — Bioinstrumentation II (laboratory instructor)

2019 - 2022

Lecturer (Full-time)

Faculty of Applied Sciences, Uva Wellassa University of Sri Lanka

- Instructor of record for eight undergraduate courses in electronics, embedded systems, and intelligent systems
- Designed curricula for seven new courses for the B.Sc. in Mechatronics curriculum revision, including Computer Vision and IoT Development
- Elected member of the Faculty Board; subject coordinator for a cross-faculty course serving three faculties
- Supervised 15 undergraduate final-year research projects, twelve of which produced peer-reviewed conference papers

2018 - 2019

Demonstrator (Temporary)

Faculty of Applied Sciences, Uva Wellassa University of Sri Lanka

- Led undergraduate laboratory and practical sessions in electronics, instrumentation, and workshop practice
- Maintained teaching laboratory equipment and assisted with demonstrations, practical assessments, and student project supervision

2017 - 2018

Research and Development Intern

Research & Innovation Department, MAS Intimates (Unichela Pvt Ltd, Vidiyal Division)

- PLC-based finished-goods scanner and garment folder systems
- Arduino-based chiller-off SMS alert system
- Pedal-less and automated sewing systems
- Waist-band cutter and mark-elimination systems for mesh and elastic attachment

Awards

2026

3rd Place, Three Minute Thesis (3MT) Competition

Institute for Integrative Precision Agriculture (IIPA), University of Georgia

Third place at the IIPA Retreat; includes a \$500 conference travel award.

2026

CENGR Graduate Student Travel Grant

College of Engineering, University of Georgia

Travel support for the ASABE Annual International Meeting, Indianapolis, Indiana.

2025

Hall of Fame Awardee

The Graduate School, Mississippi State University

College of Engineering honoree.

2024

TAGGS Travel Scholarship

The Graduate School, Mississippi State University

Travel scholarship for the ASABE Annual International Meeting.

2024

AG2PI Travel Award

Agricultural Genome to Phenome Initiative (AG2PI)

Travel award for the ASABE Annual International Meeting.

2023

SEC Emerging Scholar

Southeastern Conference / Mississippi State University

Selected to represent Mississippi State University in the SEC Emerging Scholars Program (2023–2024).

2023

Best Paper Award

SPIE — Autonomous Air and Ground Sensing Systems for Agricultural Optimization and Phenotyping VIII

For 'Multi-ripeness level blackberry detection using YOLOv7 for soft robotic harvesting'.

Teaching

2024

Graduate Teaching Assistant — ABE 4423/6423 Bioinstrumentation II

Mississippi State University

2022

Subject Coordinator — BGE 121-2 Ethics and Law Basics

Uva Wellassa University

- Cross-faculty course delivered to three faculties; online delivery for up to 500 students
- Coordinated visiting resource persons, assessment, and results submission

2019 - 2022

Instructor of Record — 8 undergraduate courses (18 credit hours)

Uva Wellassa University of Sri Lanka

- SCT 364-2 Embedded Systems (2019, 2020, 2021, 2022)
- SCT 462-2 Intelligent Systems (2019, 2020, 2021, 2022)
- SCT 319-2 Maintenance, Production, and Project Management (2021, 2022)
- SCT 363-2 Electrical Systems (2020, 2021)
- SCT 408-2 Industrial Electronics (2020, 2021)
- SCT 362-3 Digital and Analog Electronics (2020)
- SCT 373-2 Electric Power and Machines (2019)
- SCT 142-3 Engineering Workshop (2019)

2019

New Course Development — 7 modules

BSc in Mechatronics, Uva Wellassa University

- Designed seven new course modules and lesson plans for the programme's curriculum revision
- MEC 361-2 Computer Vision; MEC 331-2 IoT Development; MEC 232-2 Embedded Systems; MEC 233-2 Embedded Systems Design
- MEC 111-2 and MEC 212-2 Computer Programming I & II; MEC 362-2 Transducers and Instrumentation

Courses Prepared to Teach

Introduction to Robotics and Mechatronics · Computer Vision for Agricultural and Biological Systems · Artificial Intelligence for Agricultural and Biological Engineering · Deep Learning for Agricultural Robotics and Autonomous Systems (graduate) · IoT and Embedded Systems for Smart Agriculture · Sensors and Bioinstrumentation · Machine Learning Applications in Agriculture · Agricultural Automation and Precision Agriculture · Intelligent Systems

Publications

ORCID 0009-0007-2576-4348

ORCID

Google Scholar profile

Google Scholar

Citation record and metrics.

Publications

This website

Journal articles, conference proceedings, and book chapters with abstracts and links.

Conference Presentations

2026

Harvesting cotton before it's too late: a smarter robotic approach

UGA Institute for Integrative Precision Agriculture Spring Retreat, Griffin, GA

Third Place, Three Minute Thesis oral presentation.

2026

Develop and evaluate a simplified end-effector for robotic selective cotton harvesting

ASABE Annual International Meeting, Indianapolis, IN

2025

Perception-enabled manipulator control for a robotic cotton picker with dual-side harvesting capability

8th IFAC Conference on Sensing, Control and Automation Technologies for Agriculture (AGRICONTROL 2025), Davis, CA

2025

Cotton-SAM: AI-driven cotton boll localization for autonomous picking using YOLO and SAM integration

AI in Agriculture and Natural Resources Conference, Starkville, MS

2024

Which is better cotton picker? (Three Minute Thesis)

Mississippi Agriculture Consortium, Mississippi State University

2024

Development of a cotton boll detection system to enhance autonomous picking using YOLOv8 and SAM

ASABE Annual International Meeting, Anaheim, CA

2024

Cotton boll localization system to enable autonomous cotton picking using YOLO and SAM

Graduate Student Research Symposium, Mississippi State University

2024

Computer vision-enabled autonomous robotic navigation system for cotton farms in a Gazebo simulation environment

ASABE Annual International Meeting, Anaheim, CA

2023

Automating catfish cutting process using deep learning-based semantic segmentation

SPIE Defense + Commercial Sensing, Orlando, FL

2023

Catfish cutting automation through computer vision and deep learning

Mississippi Academy of Sciences 5th Summer Science & Engineering Symposium

2023

Multi-ripeness level blackberry detection using YOLOv7 for soft robotic harvesting

SPIE Defense + Commercial Sensing, Orlando, FL

Best Paper Award — Autonomous Air and Ground Sensing Systems for Agricultural Optimization and Phenotyping VIII session.

2018

Research Projects

2023 - PRESENT

Robotic cotton harvesting (Ph.D. research)

- CottonSim: open-source ROS/Gazebo simulator and virtual cotton farm for autonomous navigation testing
- Cotton-Eye perception module using RGB-D sensing and YOLOv8n-seg instance segmentation (85.2% mAP)
- Manipulator control for dual-side selective picking on a UR5e
- Selective cotton boll localization: benchmarked detection and segmentation models under field conditions, validated on a robotic manipulator (arXiv:2609.19592)
- Technology disclosure filed with the MSU Office of Technology Management (2025); co-inventor with Dr. Xin Zhang

2023 - PRESENT

Research software and data

- CottonID (2026): web application for cotton boll detection, yield estimation, and defoliation-timing decision support (Next.js, FastAPI, RF-DETR/YOLO)
- CottonSim (2025): open-source ROS/Gazebo simulator and vision-guided navigation stack — github.com/imtheva/CottonSim
- Multi-Ripeness Blackberry (2025): detection models and annotated field dataset (1,086 images, three ripeness classes) — github.com/Zhanglab-abe/Multi-Ripeness_Blackberry
- Catfish-Segment (2023): semantic segmentation models for automated catfish processing — github.com/Zhanglab-abe/Catfish-Segment

2022 - 2025

Perception for specialty crops and food processing

- Multi-ripeness blackberry detection for soft robotic harvesting (nine YOLO architectures, 1,086 field images)
- Semantic segmentation for automating the catfish cutting process (SegFormer-B5, 89.2% mIoU)

2019 - 2022

Undergraduate research supervised at Uva Wellassa University

- Traffic sign recognition and auditory alert system for Sri Lankan drivers (IEEE SCSE 2024)
- Automated image capture for the zebrafish embryo toxicity model
- IoT-based automated colony counter
- Virtual reality for robot manipulator education (IEEE MERCon 2023)
- Sinhala printed-text recognition for postal automation
- Automatic billing system for a Sri Lankan pastry shop

Service, Mentoring & Professional Development

2024 - PRESENT

Undergraduate Research Mentor — 5 students

Sensing and Automation in Agri-Systems (SAAS) Lab

Mentored five undergraduate researchers across Mississippi State University and the University of Georgia, including a visiting SUROE summer researcher from Trevecca Nazarene University. One ongoing.

2024 - PRESENT

Lab Manager

Sensing and Automation in Agri-Systems (SAAS) Lab

Mississippi State University (2024–2025); University of Georgia (2025–present).

2026 - 2026

Peer Review Certification — 3 courses, 14 hours

Elsevier Researcher Academy

Fundamentals of Peer Review; Certified Peer Reviewer Course; Becoming a Peer Reviewer.

2024 - 2025

Preparing Future Faculty Program

Mississippi State University

2024 - 2024

Certified Teaching Assistant (Level 2/3)

Mississippi State University

2023 - 2023

Certificate of Competency — Fundamentals of Deep Learning

NVIDIA Deep Learning Institute

2022 - 2023

Member, Faculty Board

Faculty of Applied Sciences, Uva Wellassa University

Appointed by the Dean under S.48(1A)(c) of the Universities Act No. 16 of 1978; three-year term, curtailed on departure for doctoral study.

2022 - 2023

Program Review and Self-Evaluation Report (SER) Group Member (groups 3, 4, and 8)

Uva Wellassa University

2022 - 2023

Subject Coordinator, BGE 121-2 Ethics and Law Basics

Faculty of Applied Sciences, Uva Wellassa University

Coordinated a cross-faculty course delivered to three faculties: scheduling with visiting resource persons, online delivery for up to 500 students, assessment and results submission.

2022 - 2023

Member, Graduate Employability Survey (AHEAD ELTA-ELSE Faculty Development Plan)

Faculty of Applied Sciences, Uva Wellassa University

Appointed by the Dean to conduct the faculty-wide graduate employability survey informing the final AHEAD project report.

2022 - 2023

Student Counselor, Faculty of Applied Sciences

2019 - 2023

Academic Sub-Warden

Uva Wellassa University

2019 - 2023

Student Mentor

Uva Wellassa University

2019 - 2022

Coordinator, Short Course in Robotics and Arduino Programming

Uva Wellassa University

2021 - 2022

Psychological Counselling Program – 40 hours (World Bank AHEAD Project)

Staff Development Centre, Uva Wellassa University

2022 - 2022

Track Coordinator – Engineering and Technology, International Research Conference

Uva Wellassa University

2019 - 2020

Organizing Committee Member, International Research Conference

Uva Wellassa University

2020 - 2020

Panel Member, Curriculum Revision Stakeholder Meeting

Uva Wellassa University

2020 - 2020

Panel Member, International Research Conference

Uva Wellassa University

2019 - 2020

Faculty Development Project Member (AHEAD Project, Activity 02)

Uva Wellassa University

2020 - 2020

Workshop – Writing Manuscripts and Proposals for Research Grants

Uva Wellassa University

2020 - 2020

Co-author, Postgraduate Admissions Survey

Faculty of Applied Sciences, Uva Wellassa University

Prepared a survey of postgraduate admission pathways followed by UWU graduates; presented to and acknowledged by the Faculty Board (119th meeting).

2020 - 2020

Workshop – Online Teaching Tools and Methods for Effective Teaching and Assessment

Uva Wellassa University

2019 - 2019

Certificate Course in Teaching in Higher Education

Staff Development Centre, Uva Wellassa University

2019 - 2019

Organizing Committee Member, Convocation

Uva Wellassa University

Peer Review & Service

2024 - PRESENT

Journal Reviewer — 6 reviews completed

Computers and Electronics in Agriculture (Elsevier)

2026

Journal Reviewer — 2 reviews completed

Journal of Food Composition and Analysis (Elsevier)

2026

Journal Reviewer — 1 review completed

Food and Energy Security (Wiley)

2026

Journal Reviewer — 1 review completed

Journal of Atmospheric and Solar-Terrestrial Physics (Elsevier)

2026

Journal Reviewer — 1 review completed

Food and Humanity (Elsevier)

2024 - 2025

Secretary

Softball Cricket Club, Mississippi State University

2024 - 2025

Associate Editor — Engineering / Computer Science

Endeavors: Mississippi State Undergraduate Research Journal (ISSN 3071-012X)

Inaugural editorial board of MSU's first peer-reviewed undergraduate research journal; double-blind peer review of engineering and computer science submissions for Volume 1.

2023 - 2024

Webmaster

Sri Lankan Association, Mississippi State University

2023 - 2024

Social Secretary

Softball Cricket Club, Mississippi State University

2021 - 2023

Faculty Advisor

IEEE Robotics and Automation Society Chapter, Uva Wellassa University

2021 - 2022

Reviewer

International Research Conference, Uva Wellassa University

2021

Reviewer

2nd International Conference on Advancements in Computing, SLIIT

2016 - 2018

Member

IEEE Student Branch, Uva Wellassa University

2018

Workshop Instructor — Engineering Machinery Workshop for School Students

Uva Wellassa University

2016

Organizing Committee — UWU Robot Battles

Department of Science & Technology, Uva Wellassa University

Professional Memberships

 Student Member, Integrative Precision Agriculture Student Club (IPAC)

University of Georgia

2025

 Graduate Student Member, American Society of Agricultural and Biological Engineers (ASABE)

ASABE

2024

 Student Member, SPIE — the International Society for Optics and Photonics

SPIE

2023

 Graduate Student Member, Institute of Electrical and Electronics Engineers (IEEE)

IEEE

2020

Skills

 Programming

Python

C++

MATLAB

 Deep Learning

PyTorch

TensorFlow/Keras

YOLO (v5–v26), RF-DETR, SegFormer

Instance and semantic segmentation

Dataset design and annotation

 Robotics and Perception

ROS, Gazebo, OpenCV

RGB-D and stereo vision (ZED2i, RealSense)

UR5e manipulator control

Clearpath Husky; GPS and map-based navigation; SLAM

 Hardware and Embedded

Arduino, PLC, microcontrollers

IoT sensor networks

Instrumentation design

Languages

English

Full professional proficiency

Tamil

Native

Sinhala

Professional working proficiency

Interests

Agricultural Robotics

Selective harvesting

Field robotics

Mobile manipulation

Simulation-to-real transfer

Computer Vision

Object detection

Instance segmentation

3D perception

Ripeness and quality assessment